Prenatal Neural Construction Is Unfinished at Birth

Human cortical development proceeds through overlapping proliferation, migration, and transient zones; the brain remains immature at birth and continues under postnatal experience.

GRADE: HIGH-STRUCTURE; LOW-COGNITION-FORECAST | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Human cortical development is a sequence of overlapping processes (proliferation, migration, formation of transient embryonic zones including the subplate). Construction is unfinished at birth; postnatal experience continues the work. Structure is not intelligence or personality.

EVIDENCE AND CAUSAL ASSESSMENT

Bystron, Blakemore & Rakic (2008) revise the 1970 Boulder Committee model with human evidence: ventricular/subventricular/intermediate/cortical-plate/marginal zones, and a distinct, functionally important subplate not recognized in 1970. This is developmental neuroanatomy, not an RCT of "brain building." Severe disruptions (e.g. congenital infection, extreme growth restriction) are contrast cases, not the typical claim.

EFFECT SIZE AND PERSISTENCE

No universal cognitive effect size. Persistence of a structural difference must be measured; many prenatal differences change form after birth.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (neuroembryology and cortical development, not an observational population phenotype association).

Indirect fetal imaging; postmortem and clinical-sample selection; species differences even in a human-focused review; postnatal confounding of later outcomes.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean the newborn brain is a blank slate, or that prenatal construction writes adult character.

WHAT ACTION IS REASONABLY JUSTIFIED

Protecting pregnancy against known severe disruptors is justified. "Prenatal brain training" is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Bystron, Blakemore & Rakic (2008), Nature Reviews Neuroscience